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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 10: Congruent Triangles

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - the definition of congruence, the SAS postulate, Theorems 10.1 to 10.4 with proofs, solved exercise questions, the book's MCQs with a checked answer key, and exam practice.

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> **How to use these notes:** This is a proof unit - marks come from writing **statements with reasons**, not from answers alone. Learn each theorem in the fixed exam format used in these notes: **Given, To Prove, Construction (if any), Proof**. Copy each proof once, then close the notes and write it again from the figure only.

What this unit covers

1. Congruent figures, the six elements of a triangle, and the definition of congruent triangles
2. Correspondence, the symbol for congruence, and CPCTC
3. The five congruence rules: SAS, ASA, AAS, SSS and HL - and why SSA and AAA fail
4. Theorem 10.1: AAS congruence theorem, with ASA as its corollary
5. Theorem 10.2: Converse of the isosceles triangle theorem
6. Theorem 10.3: SSS congruence theorem
7. Theorem 10.4: HL (Hypotenuse-Leg) congruence theorem
8. Solved questions from Exercise 10.1, practice with hints, Review Exercise MCQs with answers, and common mistakes

Source book: KPTB Mathematics for Grade IX, Unit 10 (book pages 233-241). The notes follow the book's order, theorem numbers and exercise questions. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition. Answers marked "book key" are taken from the book's own Answers section.

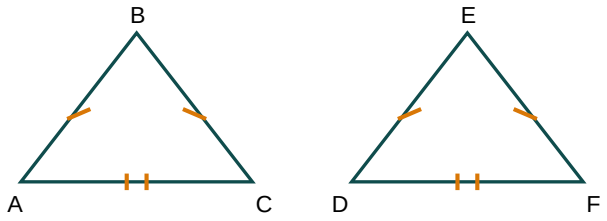
1. Congruent Figures and Congruent Triangles

Two plane figures are **congruent** if they have the **same size and the same shape**. One can always be placed exactly on top of the other so that it covers it completely - this matching is called **superposition**. Congruent figures are exact duplicates: spare parts of a machine are made congruent to the original parts so that they fit in the same place. When two figures A and B are congruent, we write **A ≅ B**. The sides and angles that match are called **corresponding** parts.

A triangle has six elements

The six elements of triangle ABC are its **three sides** AB, BC, CA and its **three angles** ∠A (∠BAC), ∠B (∠CBA) and ∠C (∠ACB). Congruence of triangles is a statement about all six elements at once.

Definition (book): Two triangles △ABC and △DEF are congruent if all six conditions hold: (i) ∠A ≅ ∠D (ii) ∠B ≅ ∠E (iii) ∠C ≅ ∠F (iv) AB ≅ DE (v) BC ≅ EF (vi) AC ≅ DF. We then write **△ABC ≅ △DEF**.



Matching tick marks show congruent sides: AB ≅ DE, BC ≅ EF, AC ≅ DF.

Correspondence - the order of the letters matters

The statement △ABC ≅ △DEF is also a **correspondence** △ABC ↔ △DEF: A matches D, B matches E and C matches F - first letter with first, second with second, third with third. From this one statement you may read off all six congruences: for example, the side joining the 2nd and 3rd letters, BC, matches the side joining the 2nd and 3rd letters of the other triangle, EF.

Properties of congruence (book)

- **Reflexive (identity):** △ABC ≅ △ABC - every triangle is congruent to itself.
- **Symmetric:** if △ABC ≅ △DEF, then △DEF ≅ △ABC.
- **Transitive:** if △ABC ≅ △DEF and △DEF ≅ △XYZ, then △ABC ≅ △XYZ.

CPCTC: Corresponding Parts of Congruent Triangles are Congruent. Once two triangles are proved congruent by a rule on the next page, any remaining matching side or angle may be claimed by writing the reason "CPCTC" (the book writes: *corresponding sides / angles of congruent triangles*). Most proof questions are two steps: first prove congruence, then use CPCTC.

2. The Five Congruence Rules

The definition asks for six matching parts, but three carefully chosen parts are enough to force the other three. This unit has **one postulate** (accepted without proof) and **four theorems** built on it:

Rule	What must match	Book status
SAS	Two **sides** and the **included angle** between them	Postulate (book, p. 234)
ASA	Two **angles** and the **included side** between them	Corollary of Theorem 10.1
AAS	Two **angles** and a **non-included side**	Theorem 10.1
SSS	All **three sides**	Theorem 10.3
HL	**Right-angled** triangles only: **hypotenuse** and one **leg**	Theorem 10.4

SAS Postulate (book): If two sides and their included angle in one triangle are congruent to two sides and their included angle in another triangle, then the two triangles are congruent. Every other rule in this unit is proved using SAS.

"Included" - the word examiners test

- The **included angle** of two sides is the angle formed where those two sides meet: ∠B is the included angle of sides AB and BC. The **included side** of two angles is the side joining their vertices: BC is the included side of ∠B and ∠C.
- SAS works only with the included angle. Two sides and a non-included angle is **SSA** - and SSA is **not** a valid rule.

Two combinations that never prove congruence

AAA gives the same shape but not the same size (the triangles are similar only). **SSA** - two sides and a non-included angle - can fit two different triangles, because the free side can land in two positions. Neither is ever a valid reason in a proof - and Review Exercise 10 tests exactly these two traps (its answers: "not sufficient" = AAA, "legitimate reason" = SAS).

How to choose a rule in a proof

1. Mark given equal sides and angles on the figure, plus hidden ones: a **common side**, **vertical opposite angles**, a **midpoint**, or **base angles of an isosceles triangle**.
2. Count what you have: 3 sides → SSS; 2 sides + the angle between them → SAS; 2 angles + any side → ASA or AAS; right angle + hypotenuse + a side → HL.
3. Write the congruence in the correct letter order, then claim the required part by CPCTC.

Theorem 10.1 - AAS (Angle-Angle-Side) Congruence Theorem

Statement (book): If two angles and a non-included side of one triangle are congruent to the corresponding angles and non-included side of another triangle, then the triangles are congruent.

Given: In $\triangle ABC \leftrightarrow \triangle DEF$: $BC \cong EF$, $\angle B \cong \angle E$ and $\angle A \cong \angle D$. **To Prove:** $\triangle ABC \cong \triangle DEF$.

Construction: Suppose AC is not congruent to DF. Take a point D′ on DF such that $AC \cong D'F$. Join D′ to E.

Proof:

Statements	Reasons
$\angle C \cong \angle F$	Third angles: $\angle A \cong \angle D$ and $\angle B \cong \angle E$ (given); angles of a triangle sum to 180°
In $\triangle ABC \leftrightarrow \triangle D'EF$: $BC \cong EF$	Given
$\angle C \cong \angle F$	Proved above
$AC \cong D'F$	Construction (supposition)
$\triangle ABC \cong \triangle D'EF$	SAS ≅ SAS
Hence $\angle B \cong \angle D'EF$	Corresponding angles of congruent triangles
But $\angle B \cong \angle DEF$	Given
So $\angle D'EF \cong \angle DEF$, forcing $ED \cong ED'$; hence D′ is D itself and $AC \cong DF$	Transitive property; the supposition was false
Now in $\triangle ABC \leftrightarrow \triangle DEF$: $BC \cong EF$, $\angle C \cong \angle F$, $AC \cong DF$	Given / proved / given
$\triangle ABC \cong \triangle DEF$	**SAS ≅ SAS**

The idea to remember: AAS is proved by *reducing it to SAS* - first the third angles are equal (angle sum), then the temporary point D′ shows the second side must match as well, and SAS finishes.

Corollary - ASA (Angle-Side-Angle): If two angles and their included side in one triangle are congruent to two angles and their included side in another triangle, then the two triangles are congruent. ASA follows at once from AAS, because the third angles are equal by the angle-sum property, so the included side becomes a non-included side for the other pair of angles.

> **Writing tip:** In your own proofs, use AAS and ASA directly as reasons ("AAS ≅ AAS"). The construction proof above is written once, when the theorem itself is asked.

Theorem 10.2 - Converse of the Isosceles Triangle Theorem

Statement (book): If two angles of a triangle are congruent, then the sides opposite to those angles are congruent.

Given: In △ABC, ∠B ≅ ∠C. **To Prove:** AB ≅ AC (the sides opposite ∠C and ∠B).

Construction: Draw the bisector of ∠A, cutting BC at D; call its two parts ∠1 and ∠2, so ∠1 ≅ ∠2.

Proof:

Statements	Reasons
In △ABD ↔ △ACD: AD ≅ AD	Common side
∠1 ≅ ∠2	Construction (AD bisects ∠A)
∠B ≅ ∠C	Given
△ABD ≅ △ACD	AAS ≅ AAS (Theorem 10.1)
Hence AB ≅ AC	Corresponding sides of congruent triangles (CPCTC)

Supporting result - the isosceles triangle theorem itself: If two sides of a triangle are congruent, the angles opposite them are congruent (if AB ≅ AC then ∠C ≅ ∠B; these are the **base angles**). The SSS proof uses it in the form "△DFG is isosceles, so ∠3 ≅ ∠4". Theorem 10.2 is its converse: a triangle is isosceles exactly when its base angles are equal.

Worked example

Example 1. In △PQR, m∠Q = m∠R = 65°. Which sides are congruent, and what is m∠P?

By Theorem 10.2, the sides opposite the equal angles are congruent: opposite ∠Q is PR and opposite ∠R is PQ, so **PR ≅ PQ** (isosceles, vertex P), and m∠P = 180° - 65° - 65° = **50°**.

Example 2 (finding an unknown). In △ABC, ∠B ≅ ∠C, AB = 3x - 2 cm and AC = x + 6 cm. Find x and the side length.

By Theorem 10.2, AB ≅ AC, so 3x - 2 = x + 6, giving 2x = 8 and **x = 4**. Each side is 4 + 6 = **10 cm** (check: 3(4) - 2 = 10 cm).

Common mistake: Matching the wrong side to the wrong angle - the side opposite ∠B is always AC (the side that does not touch B).

Theorem 10.3 - SSS (Side-Side-Side) Congruence Theorem

Statement (book): In a correspondence of two triangles, if three sides of one triangle are congruent to the corresponding three sides of the other, the two triangles are congruent.

Given: In $\triangle ABC \leftrightarrow \triangle DEF$: $AB \cong DE$, $BC \cong EF$ and $CA \cong FD$.

To Prove: $\triangle ABC \cong \triangle DEF$.

Construction (book): Suppose BC is not shorter than either of the other two sides of $\triangle ABC$. Construct $\triangle GEF$, with G and D on opposite sides of EF, so that $GE \cong AB$ and $\angle GEF \cong \angle B$. Join G to D. In the figure, $\angle 1$, $\angle 2$ are the parts of $\angle G$ and $\angle 3$, $\angle 4$ the parts of $\angle D$ made by GD.

Proof - main steps:

Statements	Reasons
In $\triangle ABC \leftrightarrow \triangle GEF$: $BC \cong EF$; $\angle B \cong \angle GEF$; $AB \cong GE$	Given; construction; construction
$\triangle ABC \cong \triangle GEF$	SAS ≅ SAS
So $GF \cong AC$; but $DF \cong AC$, therefore $GF \cong DF$ ..... (i)	CPCTC; given; transitive property
In $\triangle DFG$: $\angle 3 \cong \angle 4$ ..... (ii)	$\triangle DFG$ is isosceles by (i); base angles are congruent
In $\triangle EGD$: $DE \cong EG$ (each is congruent to AB), so $\angle 1 \cong \angle 2$ ..... (iii)	$\triangle EGD$ is isosceles; base angles are congruent
$\angle 3 + \angle 1 \cong \angle 4 + \angle 2$, that is, $m\angle D \cong m\angle G$	By (ii) and (iii); addition of angles
In $\triangle DEF \leftrightarrow \triangle GEF$: $DE \cong GE$; $\angle D \cong \angle G$; $DF \cong GF$	Each proved above
$\triangle DEF \cong \triangle GEF$	SAS ≅ SAS
Therefore $\triangle ABC \cong \triangle DEF$	Both are congruent to $\triangle GEF$ (transitive property)

The book refers to this result as **S.S.S ≅ S.S.S**. The idea to remember: build a copy of the first triangle on the second triangle's base, join the two apexes, and let the two isosceles triangles force the vertex angles to be equal - then SAS finishes it.

Theorem 10.4 - HL (Hypotenuse-Leg) Congruence Theorem

Statement (book): If in the correspondence of two right-angled triangles, the hypotenuse and one side of one are respectively congruent to the hypotenuse and the corresponding side of the other, the triangles are congruent.

Given: In △ABC ↔ △DEF: ∠ABC ≅ ∠E (each a right angle), AC ≅ DF (hypotenuses) and BC ≅ EF (legs). **To Prove:** △ABC ≅ △DEF.

Construction: Produce AB to the point G so that BG ≅ DE. Join C to G.

Proof - main steps: (1) m∠CBG = 90°, the supplement of the right angle ∠ABC. (2) In △GBC ↔ △DEF: GB ≅ DE (construction), ∠GBC ≅ ∠DEF (each a right angle), BC ≅ EF (given), so △GBC ≅ △DEF by SAS, and GC ≅ DF by CPCTC. (3) But DF ≅ AC (given), so GC ≅ AC: △CAG is isosceles, hence ∠A ≅ ∠G. (4) In △ABC ↔ △GBC: ∠A ≅ ∠G, ∠ABC ≅ ∠GBC (right angles), AC ≅ GC, so △ABC ≅ △GBC by AAS. (5) Therefore **△ABC ≅ △DEF**, since both are congruent to △GBC. The book refers to this result as **H.L ≅ H.L**.

Worked example - spotting HL

Example 3. Two right-angled triangles have hypotenuses of 13 cm and a leg of 5 cm each. Are they congruent? What is the third side?

Yes - hypotenuse and one leg match, so the triangles are congruent by **HL**, whatever position the leg is drawn in. The third side is the same in both: by Pythagoras (Unit 15), $\sqrt{(13^2 - 5^2)} = \sqrt{(169 - 25)} = \sqrt{144} =$ **12 cm**. HL works because the right angle plus Pythagoras leaves no freedom: once the hypotenuse and one leg are fixed, the third side is forced.

Why HL is not "SSA in disguise"

HL looks like two sides and a non-included angle, which is exactly the SSA pattern that fails in general. It works here only because the non-included angle is a **right angle**: with a 90° angle, the hypotenuse is fixed as the longest side, the swinging-side ambiguity disappears, and Pythagoras forces the remaining side. That is why the book gives right-angled triangles their own theorem - and why HL must never be quoted for triangles without a right angle.

> **All four theorems, one line each:** AAS - two angles and a side (p. 4). Theorem 10.2 - equal angles stand opposite equal sides (p. 5). SSS - three sides (p. 6). HL - right angle, hypotenuse and leg (this page). Together with the SAS postulate and its ASA corollary, these are the only reasons you may write in this unit.

3. Solved Questions - Exercise 10.1

Q3 (book). Given: C is the midpoint of BE, and ∠B ≅ ∠E. Prove: △ABC ≅ △DEC.

Proof: In △ABC ↔ △DEC:

- BC ≅ EC - C is the midpoint of BE (given), so it divides BE into two congruent parts.
- ∠B ≅ ∠E - given.
- ∠ACB ≅ ∠DCE - vertically opposite angles, where AD and BE cross at C.

Therefore **△ABC ≅ △DEC by AAS ≅ AAS** (two angles and the non-included side BC / EC). Note the letter order: A↔D, B↔E, C↔C.

Q7 (book). Given: AB ⊥ BC, AD ⊥ DC, and AB ≅ AD. Prove: △ABC ≅ △ADC.

Proof: Both triangles are right-angled: ∠ABC = 90° because AB ⊥ BC, and ∠ADC = 90° because AD ⊥ DC. In the right-angled triangles ABC and ADC:

- AC ≅ AC - common hypotenuse.
- AB ≅ AD - given leg.

Therefore **△ABC ≅ △ADC by HL ≅ HL** (Theorem 10.4). This is the standard HL pattern: right angle, common hypotenuse, one given leg.

Q8 (book). QUAD is a rectangle. QC = x and DC = 3x - 8. Find x.

Solution: In a rectangle the diagonals are congruent and bisect each other, so every half-diagonal is congruent to every other: QC (half of one diagonal) and DC (half of the other diagonal, from vertex D to the centre C) are congruent. Hence x = 3x - 8, so 2x = 8 and **x = 4**. (Book key: x = 4.) Check: QC = 4 and DC = 3(4) - 8 = 4 - the two halves agree, as they must.

> **Pattern check:** Q3 needed a "hidden" fact twice - the midpoint gives a side, and crossing lines give vertical angles. Q7 needed the perpendicular signs turned into right angles before HL could be used. In this unit, always translate every symbol in the "Given" line (midpoint, ⊥, bisector) into a congruence before you choose a rule.

4. Exercise 10.1 - Practice With Hints (Book Questions)

1. **Prove that the perpendiculars drawn from the endpoints of the base of an isosceles triangle to their opposite sides are congruent.**
   Hint: Let the triangle be ABC with AB ≅ AC and base BC; the perpendiculars from B and C meet the opposite sides at E and D. In △BEC and △CDB: right angles at E and D, ∠ECB ≅ ∠DBC (the base angles themselves), and BC is the common hypotenuse - AAS, then the perpendiculars are congruent by CPCTC.
2. **In the given figure AC ≅ CE and ∠B ≅ ∠D. Prove that BC ≅ CD.**
   Hint: The lines cross at C, so ∠ACB ≅ ∠ECD (vertically opposite). With ∠B ≅ ∠D and AC ≅ CE, AAS gives △ABC ≅ △EDC; BC ≅ CD follows by CPCTC.
3. **C is the midpoint of BE, ∠B ≅ ∠E. Prove △ABC ≅ △DEC.** (Solved in full on page 8.)
4. **Prove that the median from the vertex angle of an isosceles triangle is perpendicular to the base and bisects the vertex angle.**
   Hint: Median means the midpoint of the base: two equal segments, the equal sides of the isosceles triangle, and the median itself common - SSS. Then CPCTC gives the bisected angle, and the two angles at the foot are equal and supplementary, so each is 90°.
5. **Prove that the bisector of the vertex angle of an isosceles triangle is the perpendicular bisector of the base.**
   Hint: Equal sides, bisected angle, common side - SAS. CPCTC gives equal segments on the base (so it bisects the base) and equal supplementary angles at the foot (so it is perpendicular).
6. **PQRS is a square. X, Y and Z are the mid-points of PQ, QR and RS respectively. Prove that △PXY ≅ △SZY.**
   Hint: PX ≅ SZ (halves of the congruent sides PQ and SR). Then build the other two pairs: the small corner triangles at Q and R are congruent by SAS (right angles, legs are halves of the sides), giving XY ≅ ZY; and △PQY ≅ △SRY by SAS, giving PY ≅ SY. Now △PXY ≅ △SZY by SSS.
7. **AB ⊥ BC, AD ⊥ DC, AB ≅ AD. Prove △ABC ≅ △ADC.** (Solved in full on page 8 - HL.)
8. **QUAD is a rectangle, QC = x, DC = 3x - 8. Find x.** (Solved on page 8: x = 4, book key.)

Review Exercise 10 - proof questions (book)

2. **In △PQR, the bisectors of ∠Q and ∠R meet at O. Prove that O is equidistant from the sides of the triangle.** Hint: Drop perpendiculars from O to the three sides; use AAS on the pairs of right triangles sharing OQ and OR - a point on an angle bisector is equidistant from the sides of the angle.
3. **In the given figure AB ≅ CB and ∠A ≅ ∠C. Prove that AE ≅ CD.** Hint: △ABE and △CBD share ∠B (common). With AB ≅ CB and ∠A ≅ ∠C, ASA/AAS gives the congruence; AE ≅ CD by CPCTC.
4. **ABC is a triangle with m∠A = 35° and m∠B = 100°, BD ⊥ AC. Prove that △BDC is an isosceles triangle.** Hint: m∠C = 180° - 35° - 100° = 45°. In right-angled △BDC, m∠DBC = 180° - 90° - 45° = 45° = m∠C, so by Theorem 10.2 the opposite sides are congruent: BD ≅ CD.

5. Solved Proof - Review Exercise 10, Q5

Q5 (book). ABC is an isosceles triangle. D is the mid-point of base BC. Prove that AD bisects ∠A and AD ⊥ BC.

Proof: In △ABD ↔ △ACD:

- AB ≅ AC - sides of the isosceles triangle (given).
- BD ≅ CD - D is the midpoint of BC (given).
- AD ≅ AD - common side.

So **△ABD ≅ △ACD by SSS ≅ SSS**. Now use CPCTC twice:

- ∠BAD ≅ ∠CAD (corresponding angles) - so **AD bisects ∠A**.
- ∠ADB ≅ ∠ADC (corresponding angles). But these two angles form a linear pair on BC, so ∠ADB + ∠ADC = 180°. Two equal angles summing to 180° are each 90° - so **AD ⊥ BC**.

This one proof is the model for Exercise 10.1 Q4 and Q5 as well: in an isosceles triangle, the median, the angle bisector and the perpendicular from the vertex all land on the same line.

Review Exercise 10 - MCQs (book) with checked answers

1. Which of the following is **not** a sufficient condition for the congruency of two triangles? a) ASA ≅ ASA b) HS ≅ HS c) SAA ≅ SAA d) AAA ≅ AAA
   Answer: (d) - three angles fix the shape, not the size.
2. The diagonal of a _____ does **not** divide it into two congruent triangles. a) rectangle b) square c) parallelogram d) trapezium
   Answer: (d) - in a trapezium the two parts have different side lengths.
3. In △ABC ≅ △DEF, which is **not** correct? a) ∠B ≅ ∠E b) CA ≅ FD c) ∠CBA ≅ ∠FED d) ∠ABC ≅ ∠EFD
   Answer: (d) - ∠ABC is the angle at B, which corresponds to E, not to F (EFD is the angle at F).
4. In △ABC, if ∠A ≅ ∠B, then the bisector of _____ divides △ABC into two congruent triangles. a) ∠A b) ∠B c) ∠C d) any one of its angles
   Answer: (c) - ∠A ≅ ∠B means the opposite sides BC ≅ AC (Theorem 10.2), so C is the vertex angle; its bisector gives SAS.
5. Which is a legitimate reason for declaring two triangles congruent? a) SSA ≅ SSA b) SAS ≅ SAS c) AAA ≅ AAA d) all of these
   Answer: (b) - only SAS, the postulate.

Answer key checked against the book's own Answers section: Review Exercise 10 - (i) d, (ii) d, (iii) d, (iv) c, (v) b.

6. Unit Summary, Common Mistakes and Final Practice

Summary - the whole unit on one page

- Congruent figures have the same size and shape; one can be superposed exactly on the other. A triangle has six elements: three sides and three angles.
- △ABC ≅ △DEF means all six corresponding parts are congruent, in the letter order A↔D, B↔E, C↔F. CPCTC: corresponding parts of congruent triangles are congruent.
- **SAS (postulate):** two sides and the included angle. **ASA (corollary):** two angles and the included side. **AAS (Theorem 10.1):** two angles and a non-included side. **SSS (Theorem 10.3):** three sides. **HL (Theorem 10.4):** right-angled triangles with congruent hypotenuse and one leg.
- **Theorem 10.2:** if two angles of a triangle are congruent, the sides opposite them are congruent - and conversely, congruent sides stand opposite congruent (base) angles. An equilateral triangle has all sides and all angles equal (each angle 60°).

Common mistakes

- **Using SSA or AAA as a reason.** Neither ever proves congruence (Review MCQs 1 and 5 test exactly this).
- **Wrong letter order.** If the proof pairs A with E, the statement must be written in that order; MCQ 3 is lost on exactly this.
- **Forgetting the reason column.** A proof is statements *with* reasons - "Given", "Common", "Vertically opposite angles", "CPCTC", and the rule used.
- **Claiming CPCTC before congruence.** First prove the triangles congruent by a rule; only then take out the needed part.
- **HL on non-right triangles, and skipping the construction.** HL needs a right angle in both triangles. Theorems 10.2, 10.3 and 10.4 each begin with a construction - a missing construction loses the proof's opening mark.

Final practice - try without looking

1. State which rule proves congruence, or write "not enough information": (a) sides 5, 7, 9 cm in both; (b) angles 40°, 60°, 80° in both; (c) two sides 6 cm, 8 cm with the included angle 50° in both; (d) two sides 6 cm, 8 cm with a non-included angle 50° in both; (e) right-angled, hypotenuse 13 cm and one leg 5 cm in both.
2. In △MNO, m∠M = m∠N = 70°. Which two sides are congruent? Find m∠O.
3. Write the full proof of Theorem 10.2 from the figure only (Given, To Prove, Construction, Proof).

> **Answers:** 1. (a) SSS (b) AAA - not enough (similar only) (c) SAS (d) SSA - not enough (e) HL. 2. Sides opposite ∠M and ∠N: ON ≅ OM (Theorem 10.2); m∠O = 40°. 3. See page 5.

End of Unit 10 notes. Source: KPTB, A Textbook of Mathematics for Grade IX, Unit 10 (book pages 233-241), including Exercise 10.1, Review Exercise 10, the summary and the answer key.